

Standardizing Hybrid Automata: From
Theoretical Foundations to Real-World
Implementation Frameworks
A Pilot Literature on the Development of a Unified Framework for
Hybrid Automata Development and Simulation

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Feb 2026

Abstract

Hybrid automata are formal models for a dynamical system with discrete and continuous components [1] widely adopted across research & industry for a variety of use cases ranging from control systems to stock trading algorithms, and from robotics to cyber-physical systems a recent example of this is for solving COLREG compliant autonomy for maritime vehicles [3] where the hybrid automata is used to model the discrete decision making process of the vessel and the continuous dynamics of the vessel's movement.

This theoretical framework provides a powerful method for modeling complex systems that can solve real world problems algorithmically without the need for human intervention during the execution of the system in a way that is efficient and effective and totally transparent for the human-in-the-loop to comprehend the decision making process of the system.

While in recent years there has been significant progress in the development of Artificial Intelligence (AI) through the rise of large-scale models like OpenAI's GPT series [2] that have shown the ability to imitate human-like reasoning further blurring the lines between human and machine intelligence therefore reducing the gap between what humans and machines can do there is still issues with AI that make it difficult to deploy in real world application such as lack of transparency and explainability due to the black-box nature of decision making caused by the topology of deep-neural networks (dnn) and also an issue with the reliability of AI systems due to their sensitivity to mistakes in novel situations that they have not been trained on and undefined emergent behaviours arising from outliers in their training data.

for the reasons presented above, hybrid automaton is still one of the most powerful tools for complex system design however, the theoretical framework has been laid there seems to be a lack of standardization in converting the theoretical framework into real world applications with several papers implementing their own versions of hybrid automaton with different algorithms enabling them.

In this pilot literature review, we will explore the current state of the art in hybrid automata development and simulation to identify the current gaps in the field and propose a unified framework for hybrid automata development and simulation that can be adopted across research and industry to standardize the development process and enable more efficient and effective use of hybrid automata in real-world applications.

0.1 introduction

Hybrid automata are formal models for dynamical systems that combine discrete modes with continuous dynamics [1]. They have been widely adopted in both research and industry for modeling complex hybrid systems such as control systems, autonomous vehicles, and cyber-physical applications. For example, hybrid automata have been used to model decision-making and motion dynamics in maritime autonomy under COLREG rules. This modeling framework provides a transparent, modular way to design and analyze systems where logic transitions and physical processes interact.

In recent years, machine learning (ML) and large AI models (e.g. GPT-4) have demonstrated remarkable capabilities on many tasks, but they often act as “black boxes” with limited interpretability. In contrast, hybrid automata yield inherently explainable models (states and transitions are explicit) and can provide formal guarantees on behavior. Consequently, hybrid automata remain a powerful tool for safety-critical and complex system design. However, despite a rich theoretical foundation, there is a lack of standardization in how hybrid automata are implemented and deployed. Different research groups and industries often develop their own modeling frameworks or code libraries, without a common API or data format. In this review, we survey the state-of-the-art in hybrid automaton development and simulation to identify gaps and motivate the need for a unified framework.

0.2 background

Hybrid automata are extensions of finite automata therefore if you are familiar with them you will notice many similarities between the two, however, the key difference being the incorporation of continuous variables and dynamics as well as resets on transitions that effect auxiliary states.

The syntax for modelling a hybrid automaton is as follows...

$$H = (Q, X, f, Init, Inv, E, G, R)$$

where:

- Q : a finite set of discrete states
- X : a finite set of continuous variables
- f : a function defining the continuous dynamics in each discrete state
- $Init$: a set of initial conditions
- Inv : a set of invariants for each discrete state
- E : a set of edges defining transitions
- G : guards enabling transitions
- R : reset maps updating continuous variables during transitions

Figure 1: Syntax for modelling a hybrid automaton

A simple example for reference is a thermostat system, to show the above syntax we define the definition of a thermostat system utilizing hybrid automaton as follows:

$$H = (Q, X, f, Init, Inv, E, G, R)$$

where:

- $Q = \{ON, OFF\}$: the discrete states representing the thermostat being on or off
- $X = \{T\}$: the continuous variable representing the temperature
- f : the dynamics defined as $f(ON) = -k(T - T_{set})$ and $f(OFF) = k(T_{ambient} - T)$ where k is a constant and T_{set} and $T_{ambient}$ are the set-point and ambient temperatures respectively
- $Init$: the initial conditions could be defined as $Init = \{(OFF, T_0)\}$ where T_0 is the initial temperature
- Inv : the invariants could be defined as $Inv(ON) = \{T < T_{set}\}$ and $Inv(OFF) = \{T > T_{ambient}\}$
- E : the edges defining transitions could be defined as $E = \{(OFF, ON), (ON, OFF)\}$
- G : the guards enabling transitions could be defined as $G(OFF, ON) = \{T < T_{set}\}$ and $G(ON, OFF) = \{T > T_{ambient}\}$
- R : the reset maps updating continuous variables during transitions could be defined as $R(OFF, ON) = T$ and $R(ON, OFF) = T$

Figure 2: Example of a hybrid automaton for a thermostat system

0.3 State of the art

0.4 Discussion / Reserach Gaps

Currently there is alot of work on automaton, but there is a lack of stardaization in bridgin the gap between the theoricial definition and the actual implementation of the autoamton in the real world. Projects like [?] however they arent really focused on stardization of development of hybrid automaton and are quite difficult to use, Therefore there is a need for a unified framework for hybrid automata development and simulation that can be adopted across research and industry to standardize the development process and enable more efficient and effective use of hybrid automata in real-world applications.

0.5 conclusions / Future Work

The gap in research/industry prevented outlines the need for a unified framework for automaton Development, therefore I propose the development of a unified framework and definition langauge for programatically converting automaton to real world application that can be adopted across research and industry. I propose that initially to make this a simple frameowrk anyone can use to create a python-based framework for simulating and running automata in real-time and offline settings. It should provide a lightweight, flexible foundation for defining & evaluating custom automata whether that be for research of real world application, whether is be a simple thermostate system or an autonomy stack. It should not only be able to simulate the automaton but also provide tools for visualizing the automaton like in figureand also provide tools for evaluating the automaton in real world settings with real world data.

blah blah blah

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